AI and Visualization

(364.090)

Finance Project · Phase I progress report

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Agenda

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03 Task

Use the **FinBERT** model to run a **sentiment analysis** and find **patterns of misclassification / bias**

- **Bearish**
- **Neutral**
- **Bullish**

04 FinBERT model

A pre-trained NLP model for sentiment analysis of financial text. Built by further training BERT on a large financial corpus and fine-tuning it for financial sentiment classification.

Financial domain training

Further trained on a large financial corpus to learn the language and terminology of finance.

Three-class sentiment

Classifies text as Positive, Negative or Neutral, with softmax probabilities.

Easy integration

Available via the Hugging Face hub.

Model output: softmax over 3 classes (example for 3 tweets)

```
['positive', 'neutral', 'negative']
[[0.89836127 0.06716516 0.03447358]
 [0.89602172 0.06137564 0.04260271]
 [0.94580281 0.0288094  0.02538782]]
```

```
BertForSequenceClassification(
  (bert): BertModel(
    (embeddings): BertEmbeddings(
      (word_embeddings): Embedding(30522, 768, padding_idx=0)
      (position_embeddings): Embedding(512, 768)
      (token_type_embeddings): Embedding(2, 768)
      (LayerNorm): LayerNorm((768,), eps=1e-12, elementwise_affine=Tru
      (dropout): Dropout(p=0.1, inplace=False)
    )
    (encoder): BertEncoder(
      (layer): ModuleList(
        (0-11): 12 x BertLayer(
          (attention): BertAttention(
            (self): BertSelfAttention(
              (query): Linear(in_features=768, out_features=768, bias=
              (key): Linear(in_features=768, out_features=768, bias=Tr
              (value): Linear(in_features=768, out_features=768, bias=
              (dropout): Dropout(p=0.1, inplace=False)
            )
            (output): BertSelfOutput(
              (dense): Linear(in_features=768, out_features=768, bias=
              (LayerNorm): LayerNorm((768,), eps=1e-12, elementwise_af
              (dropout): Dropout(p=0.1, inplace=False)
            )
          )
          (intermediate): BertIntermediate(
            (dense): Linear(in_features=768, out_features=3072, bias=T
            (intermediate_act_fn): GELUActivation()
          )
          (output): BertOutput(
            (dense): Linear(in_features=3072, out_features=768, bias=T
            (LayerNorm): LayerNorm((768,), eps=1e-12, elementwise_affi
            (dropout): Dropout(p=0.1, inplace=False)
          )
        )
      )
    )
    (pooler): BertPooler(
      (dense): Linear(in_features=768, out_features=768, bias=True)
      (activation): Tanh()
    )
  )
  (dropout): Dropout(p=0.1, inplace=False)
  (classifier): Linear(in_features=768, out_features=3, bias=True)
)
```

Sources: huggingface.co/ProsusAI/finbert · finbert.org

05 Quick look into the dataset

Twitter Financial News: an English-language corpus of annotated finance-related tweets, collected via the Twitter API and used for multi-class sentiment classification.

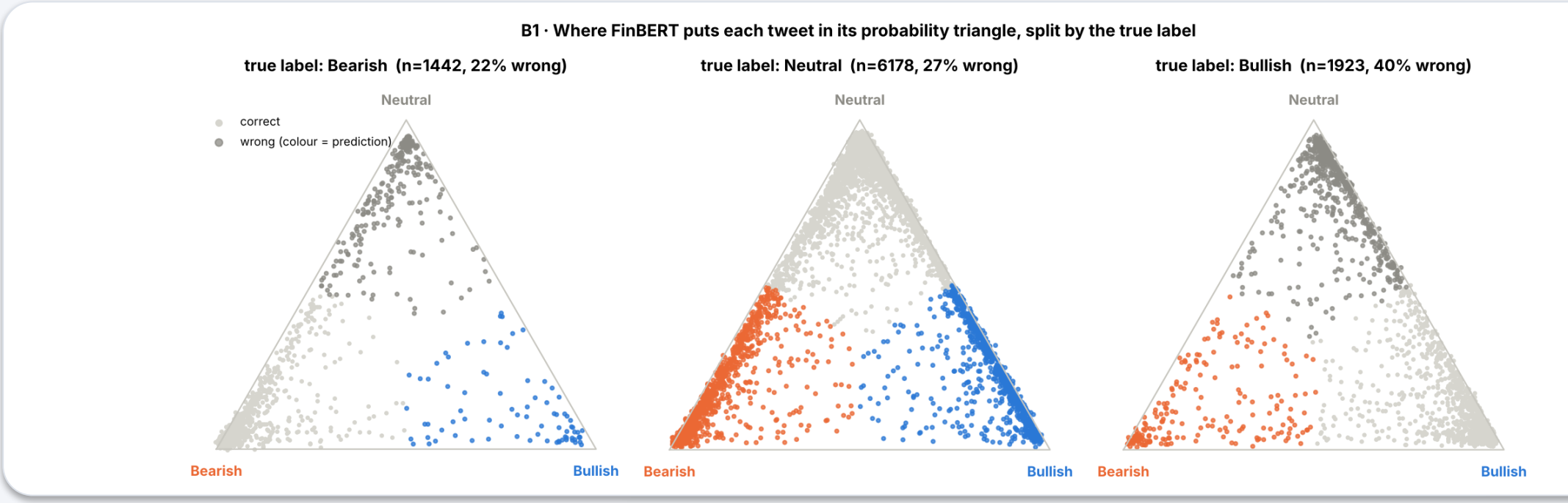
0	$BYND - JPMorgan reels in expectat
1	$CCL $RCL - Nomura points to bookir
2	$CX - Cemex cut at Credit Suisse
3	$ESS: BTIG Research cuts to Neu
4	$FNKO - Funko slides after Piper
...	
9538	The Week's Gainers and Losers on
9539	Tupperware Brands among consumer
9540	vTv Therapeutics leads healthcare
9541	WORK, XPO, PYX and AMKR among aft
9542	YNDX, I, QD and OESX amor

9543 rows × 2 columns

Label mapping (re-map for FinBERT):

LABEL_0 → Bearish | **LABEL_1 → Bullish** | **LABEL_2 → Neutral**

Source: huggingface.co/datasets/zeroshot/twitter-financial-news-sentiment



06

Metrics: Recall, Precision, Confusion Matrix

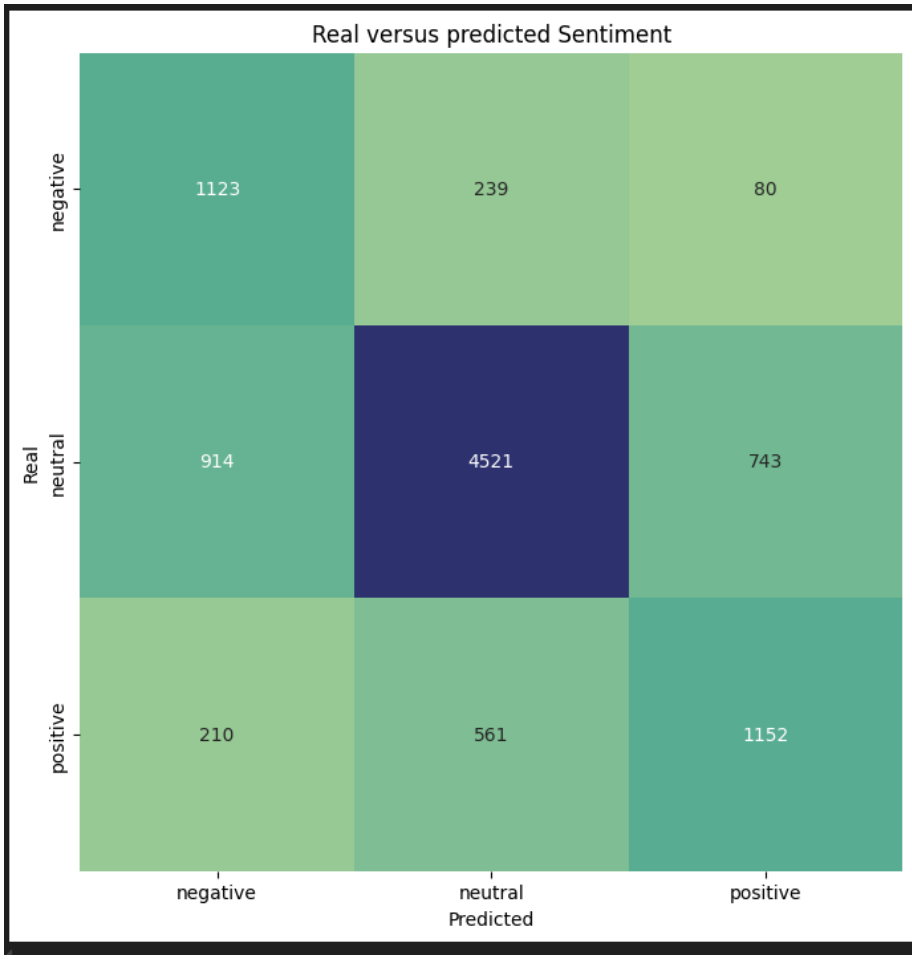
Recall

```
Pos 0.5990639625585024
Neg 0.7787794729542302
Neut 0.7317902233732599
```

Precision

```
Pos 0.5832911392405064
Neg 0.4997774810858923
Neut 0.8496523209922947
```

Recall of positive labels sticks out

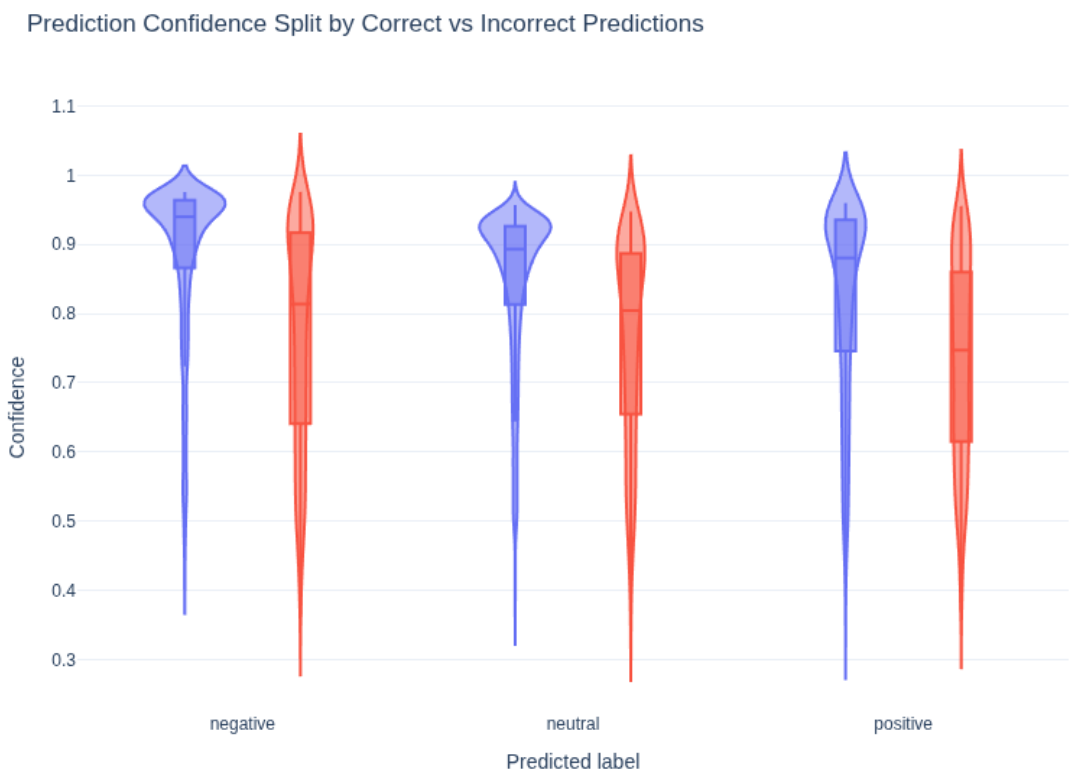


07

Observation

Kernel density estimation plot of correct vs. incorrect predictions, aggregated, shown with their confidence.

→ The model is **less confident** for incorrect predictions



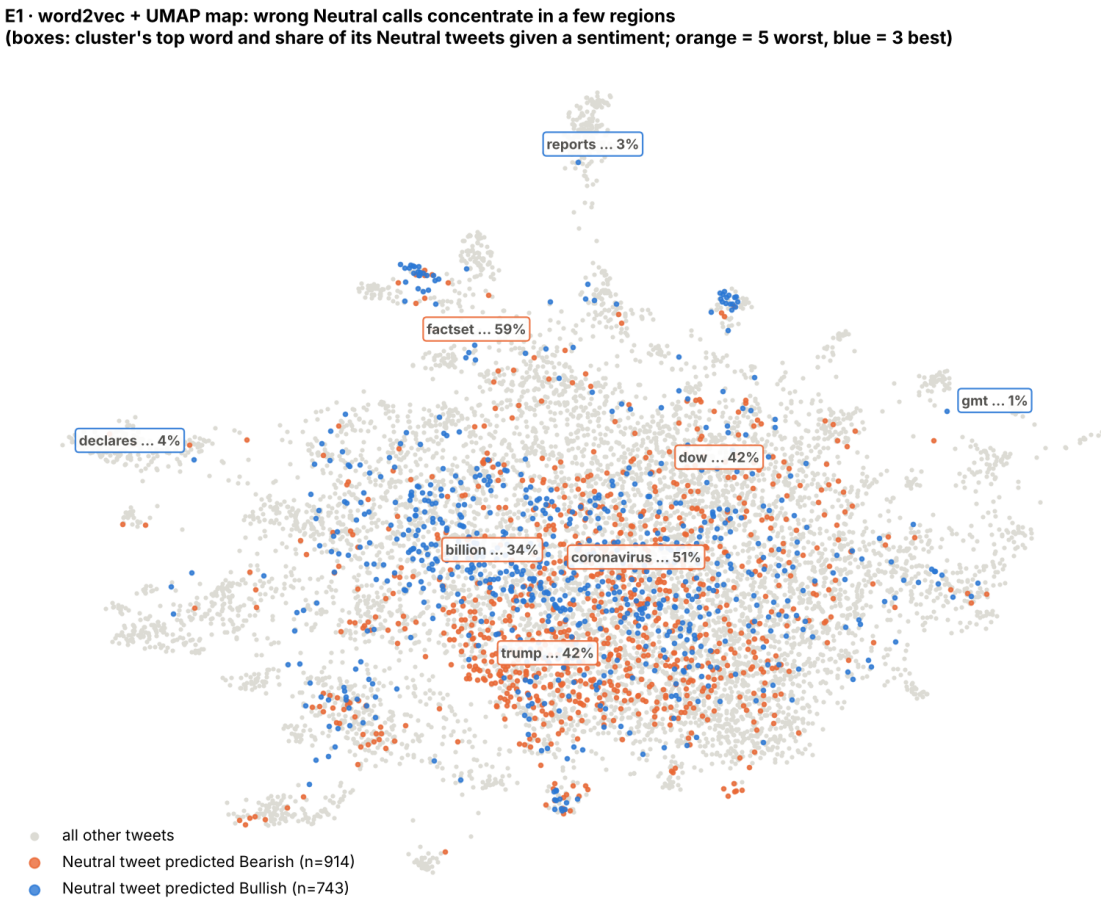
08 Word clustering

Every tweet is a dot in the vector space, placed by word2vec (gensim.models).

Only tweets with true label “neutral”. **Orange** = predicted Bearish, **Blue** = predicted Bullish.

Boxes are KMeans clusters: the 3 lowest and 5 highest error-rate clusters.

e.g. high negative (orange) misclassification in the lower middle section, about politics and viruses.

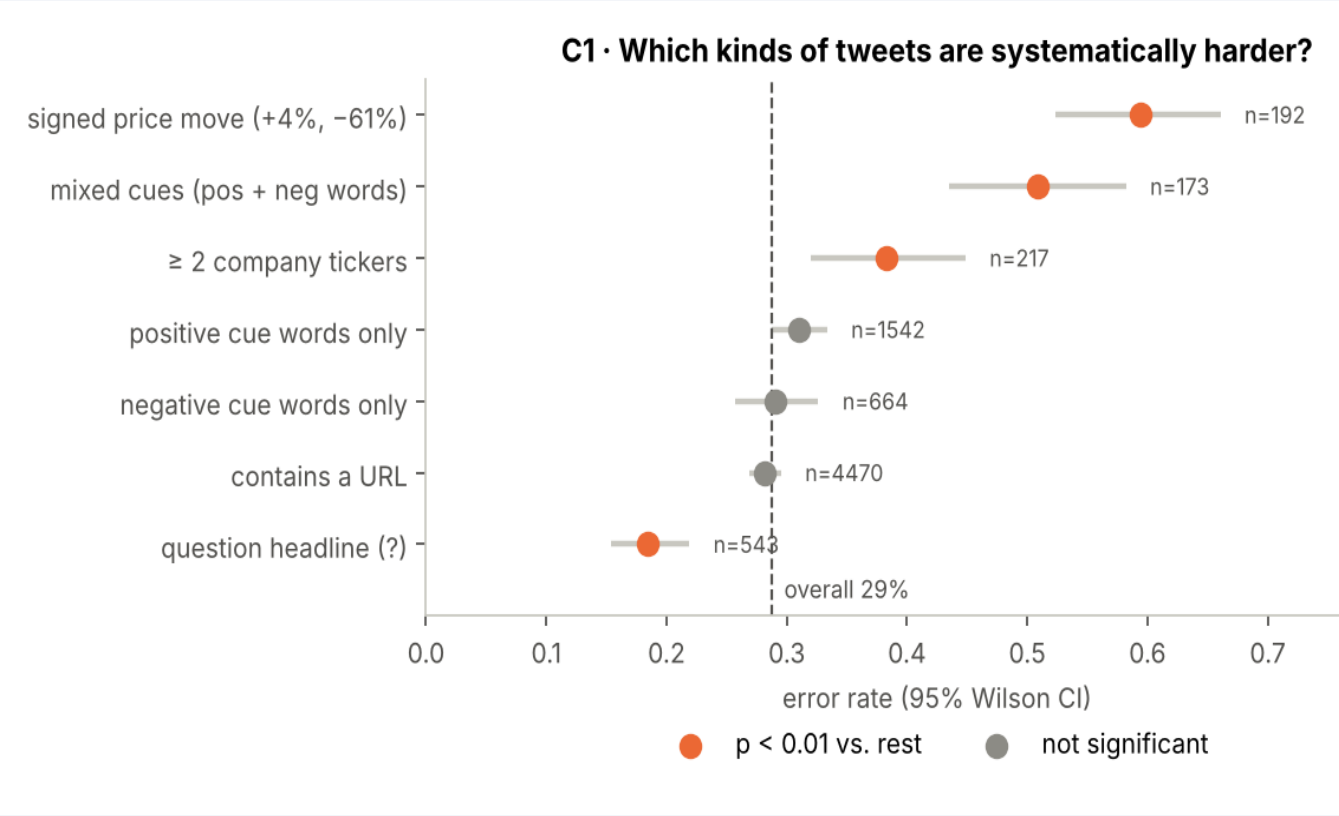


09

Other ways to systematically identify misclassifications

Looking at significant subgroups of tweets (e.g. with a price move, or the tweet contains an URL).

This plot shows **significant** subgroups (e.g. with a price move the error rate is 59%)



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Hypothesis going into Phase II

1

Investigate misclassification in 2 steps:

true label bearish → predicted bullish, and true label bullish → predicted bearish.

```
['Stock Market Update: Wholesale inventories decline in December',
 'U.S. trade deficit in goods drops to $63.2 billion in November',
 'At the end of October, housing inventory nationwide fell to a 3.9 month supply, according to NAR. A healthy market,… https:/
 'Declining Rig Counts Point To Higher Energy Prices And A Headwind For The U.S. Economy',
 'Stock Market Update: Weekly jobless claims drop below consensus']
```

Interesting,

1. since the "decline" here is interpreted as generally negative, however for the stock market it is actually positive if the inventories are more lean (implies high
2. "Deficit" probably is semantically embedded as negative, however trade deficit drop is generally positive for an economy
3. Low supply in the housing market means higher prices for housing.
4. "Declining" is probably the most important keyword here.
5. "drop" is contributing most to the models prediction

Take these tweets and investigate them in Phase 2 and look at what how the model actually classifies the individual tokens.

2

Analyse the model's reasoning

Look at these samples more deeply in Phase II.

```
['Customer Complaints Spike In FY19, Up 50% In Three Years, Says RBI Report',
 'EIA forecasts U.S. shale oil output to climb by 49,000 barrels a day in December',
 '$USO $OIL $UWT - U.S. crude supply rose 6M barrels last week, API says https://t.co
 'Stock Market Update: Market pares gains, chipmakers stay strong',
 'API data show a roughly 6 million-barrel weekly rise in U.S. crude stocks, sources
```

Take these tweets and investigate them in Phase 2 and look at what how the model actually classifies the ind